



Multi-Asset Signal Generation

DEPENDENT VARIABLES

March 4, 2020

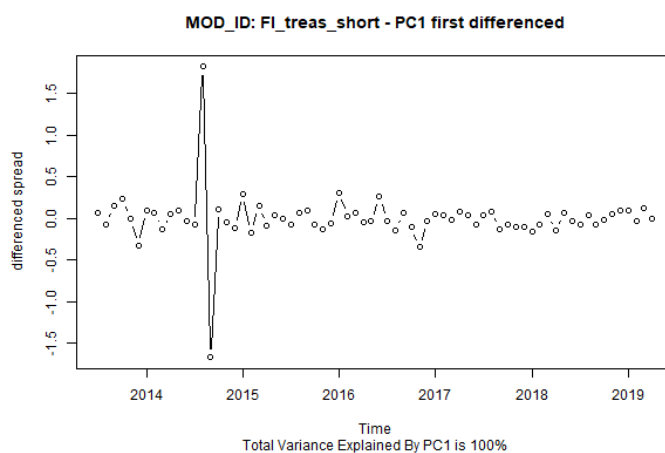
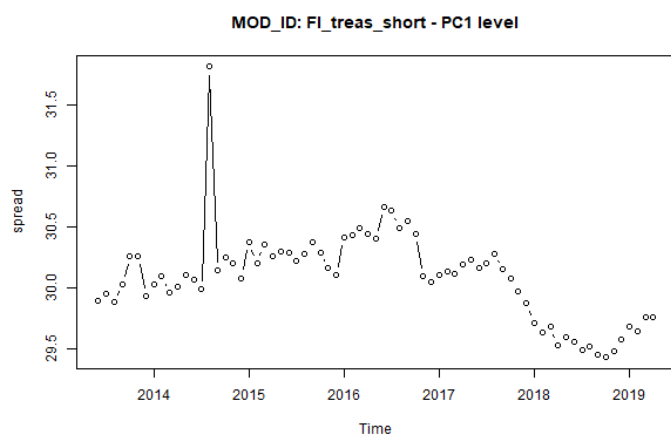
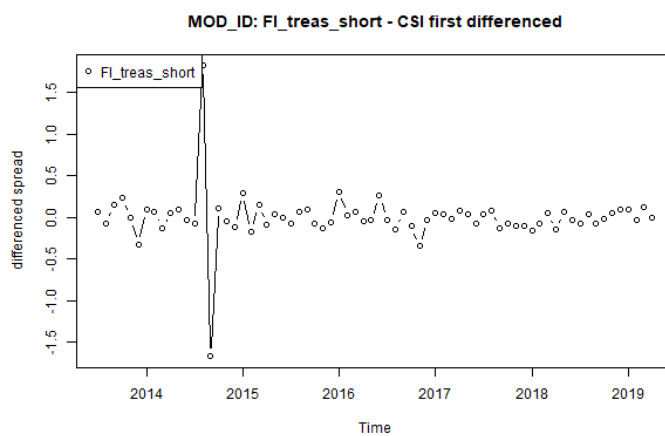
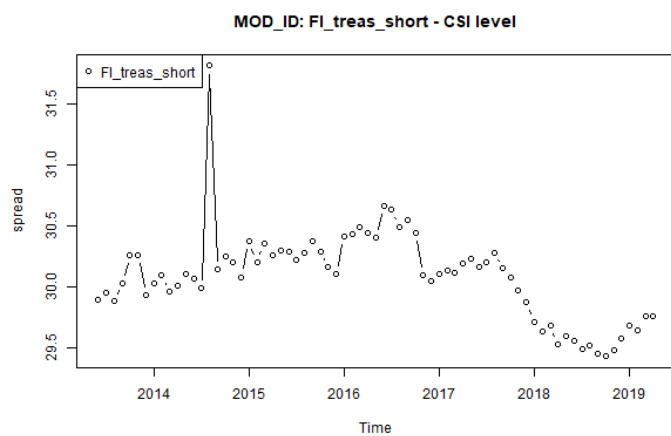
Principal Financial Group
Carnegie Mellon University

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1 PCA for Clusters

1.1 Cluster: FI_{treas_short}



	EIGENVECTOR	MULTIPLIER
FI_treas_short	1.00	1.00

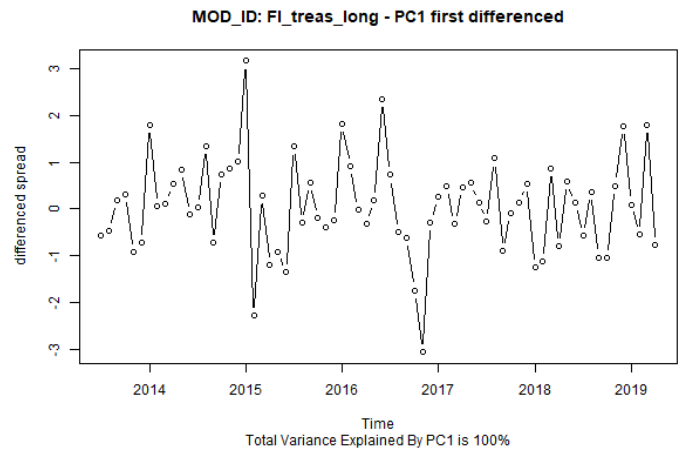
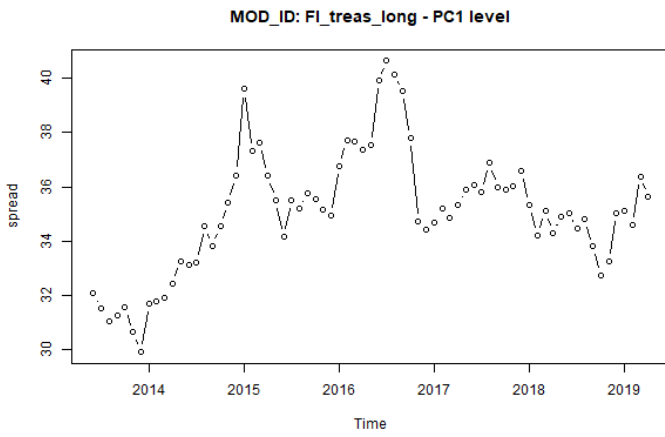
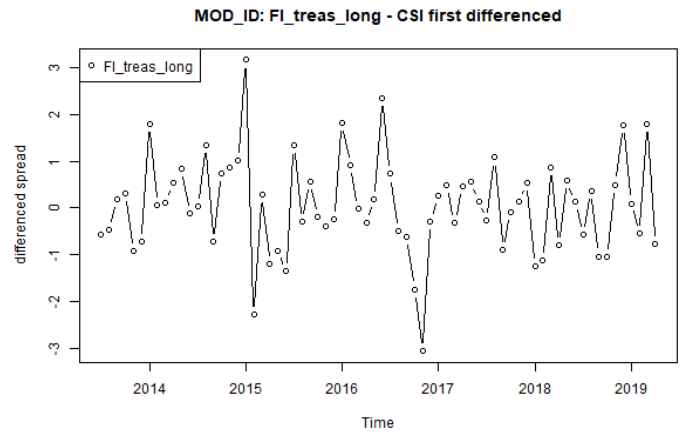
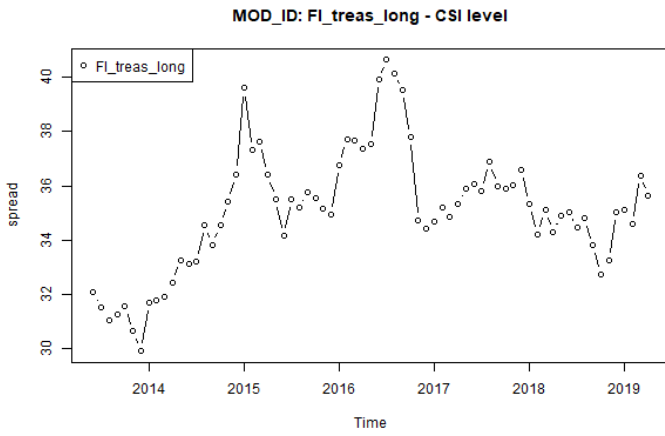
Kendall Correlation

	FI_treas_short	pc1
FI_treas_short	1	1
pc1	1	1

Spearman Correlation

	FI_treas_short	pc1
FI_treas_short	1	1
pc1	1	1

1.2 Cluster: FI_{treas_long}



	EIGENVECTOR	MULTIPLIER
FI_treas_long	1.00	1.00

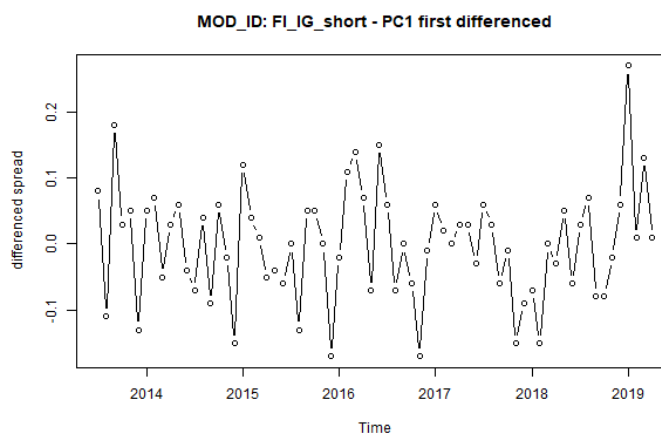
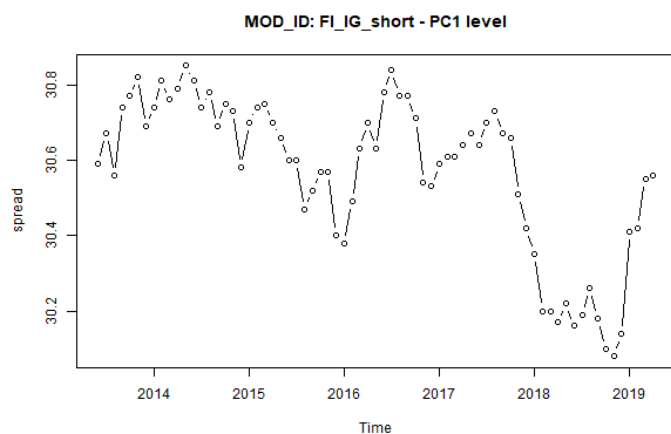
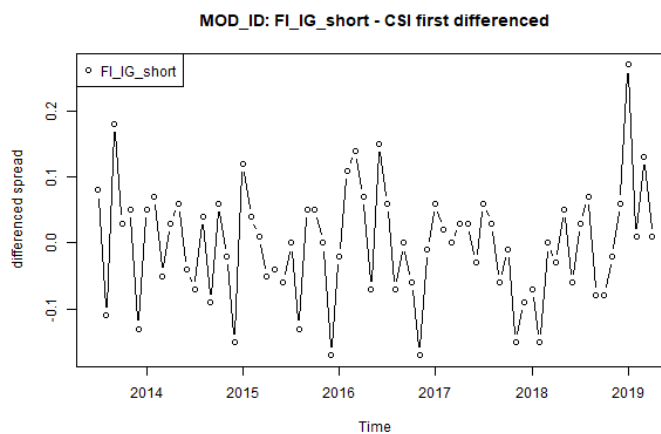
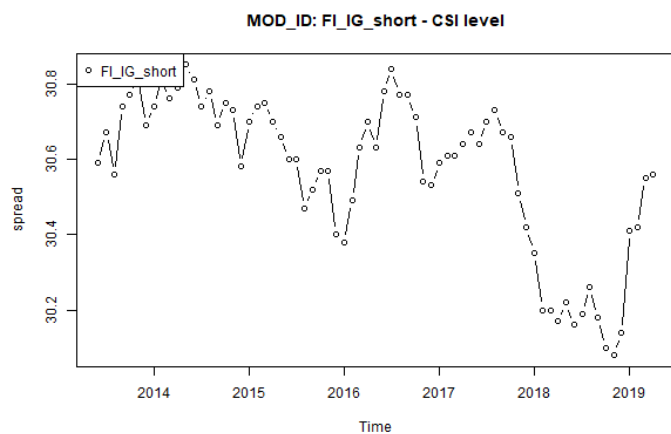
Kendall Correlation

	FI_treas_long	pc1
FI_treas_long	1	1
pc1	1	1

Spearman Correlation

	FI_treas_long	pc1
FI_treas_long	1	1
pc1	1	1

1.3 Cluster: FI_{IG_short}



Total Variance Explained By PC1 is 100%

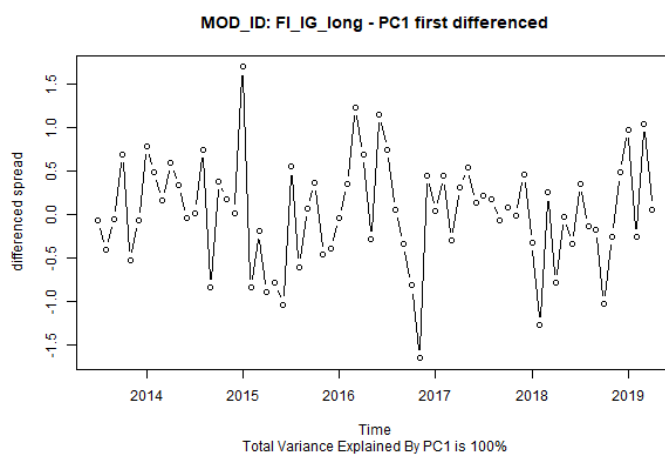
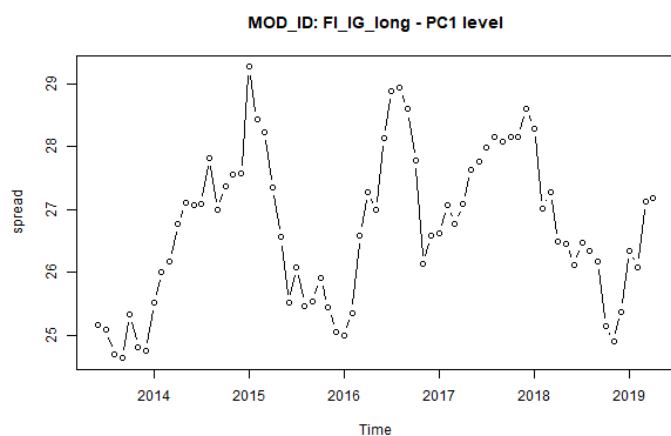
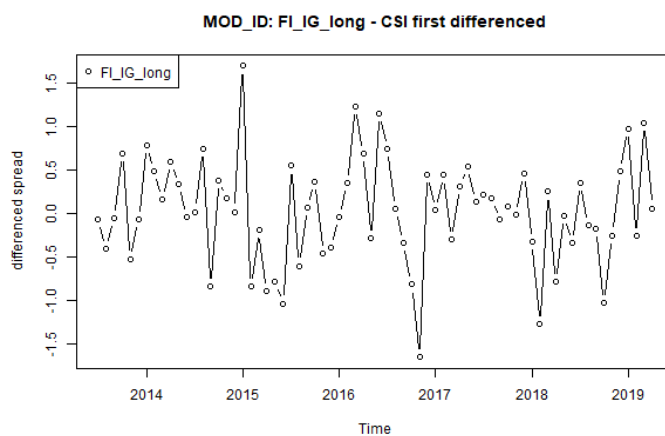
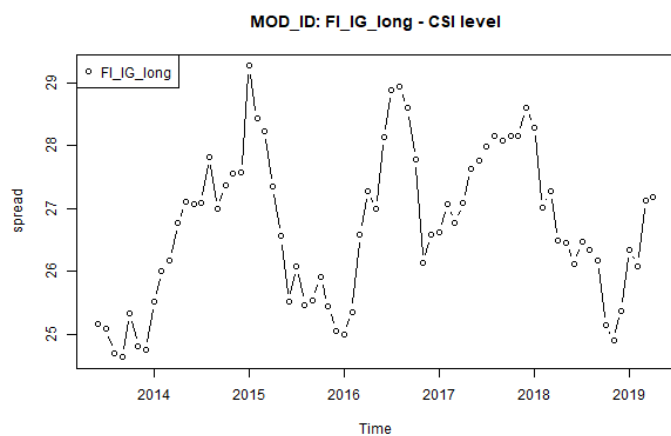
	EIGENVECTOR	MULTIPLIER
FI_IG_short	1.00	1.00

Kendall Correlation

	FI_IG_short	pc1
FI_IG_short	1	1
pc1	1	1

Spearman Correlation

	FI_IG_short	pc1
FI_IG_short	1	1
pc1	1	1

1.4 Cluster: FI_{IG_long} 

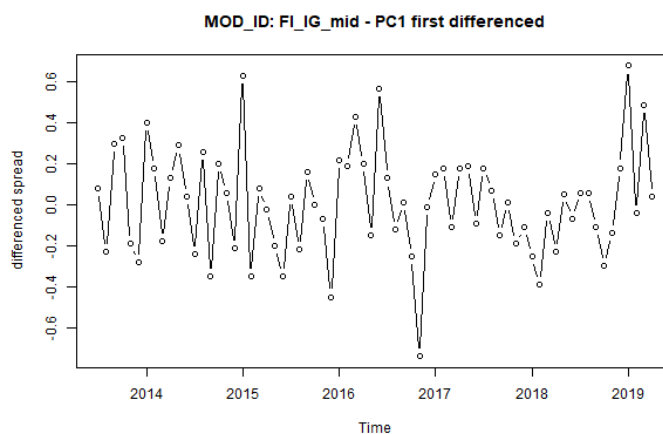
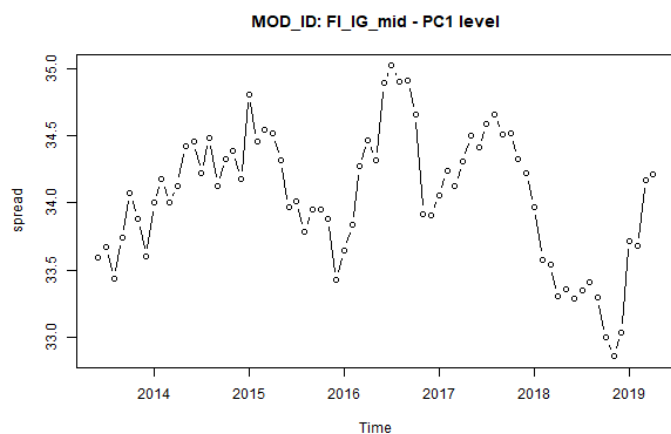
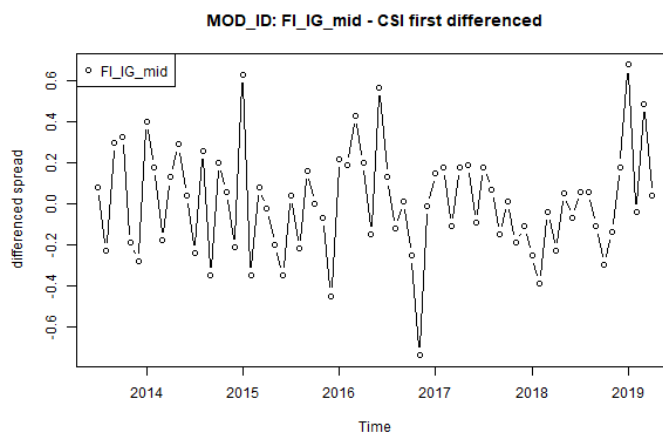
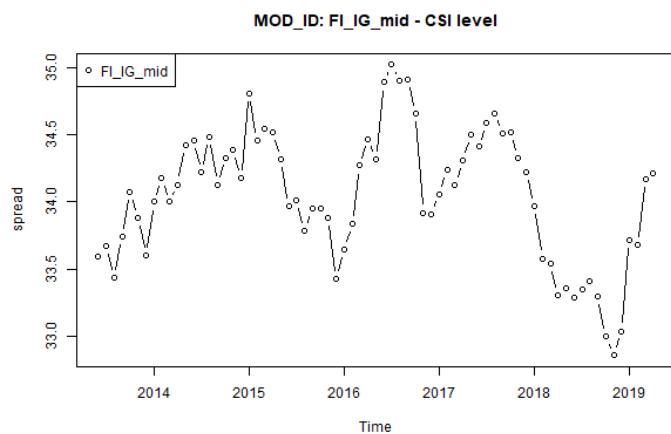
	EIGENVECTOR	MULTIPLIER
FI_IG_long	1.00	1.00

Kendall Correlation

	FI_IG_long	pc1
FI_IG_long	1	1
pc1	1	1

Spearman Correlation

	FI_IG_long	pc1
FI_IG_long	1	1
pc1	1	1

1.5 Cluster: $FI_{IG_{mid}}$ 

Total Variance Explained By PC1 is 100%

	EIGENVECTOR	MULTIPLIER
FI_IG_mid	1.00	1.00

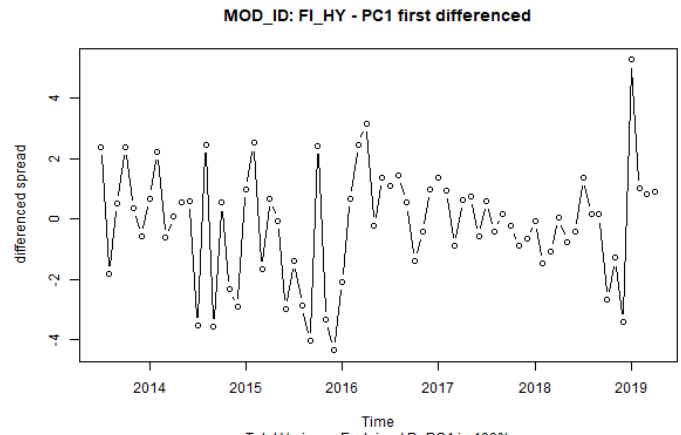
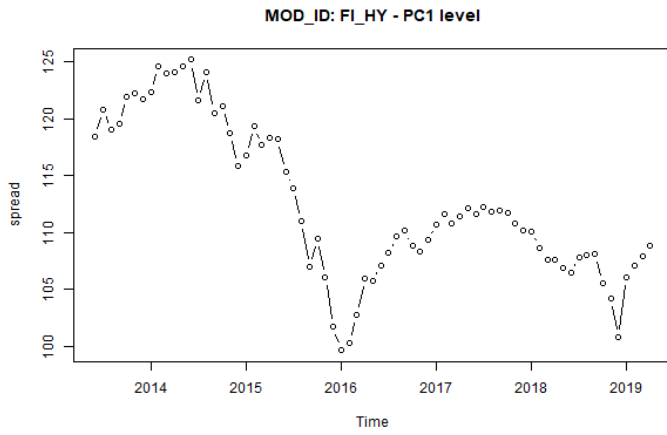
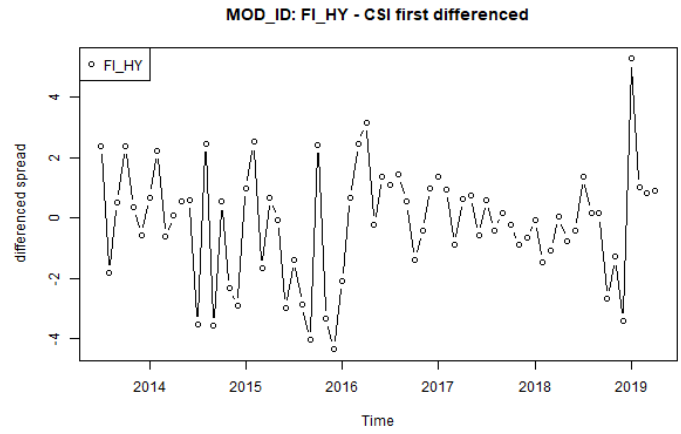
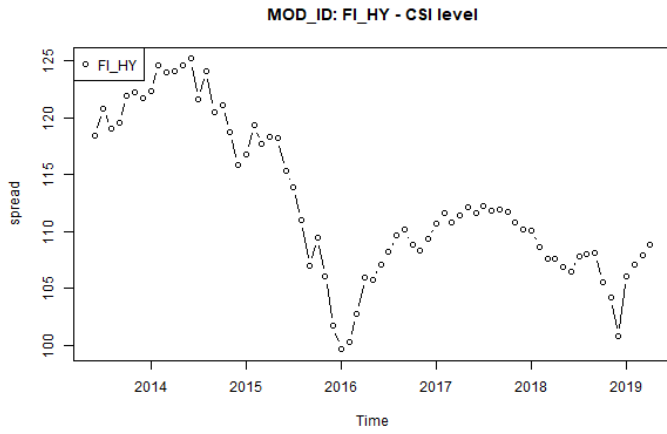
Kendall Correlation

	FI_IG_mid	pc1
FI_IG_mid	1	1
pc1	1	1

Spearman Correlation

	FI_IG_mid	pc1
FI_IG_mid	1	1
pc1	1	1

1.6 Cluster: FI_{HY}



Total Variance Explained By PC1 is 100%

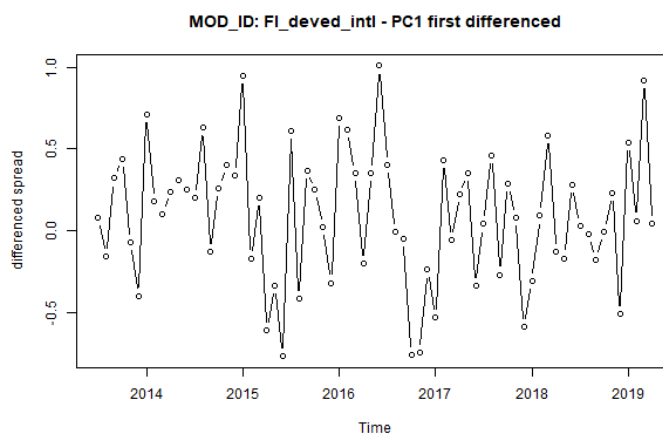
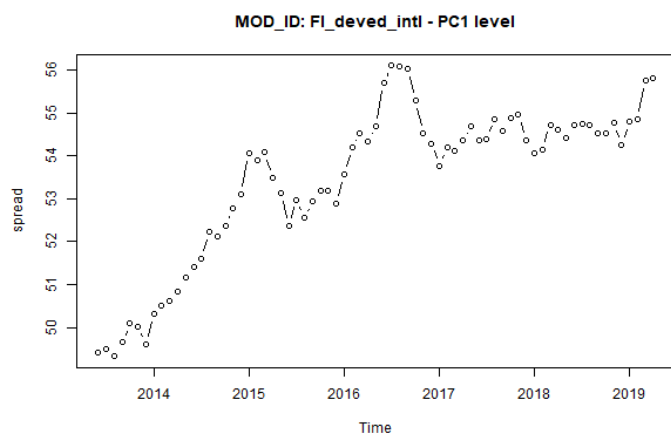
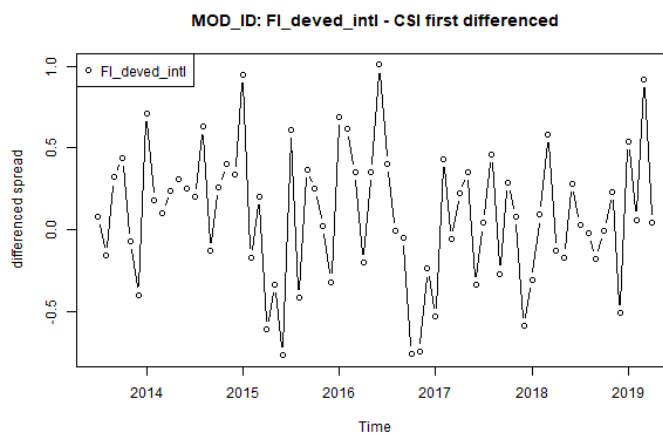
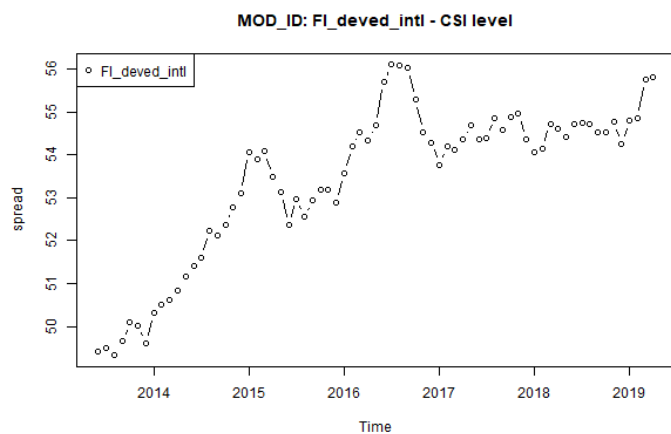
	EIGENVECTOR	MULTIPLIER
FI_HY	1.00	1.00

Kendall Correlation

	FL_HY	pc1
FL_HY	1	1
pc1	1	1

Spearman Correlation

	FL_HY	pc1
FL_HY	1	1
pc1	1	1

1.7 Cluster: FI_{deved_intl} 

Total Variance Explained By PC1 is 100%

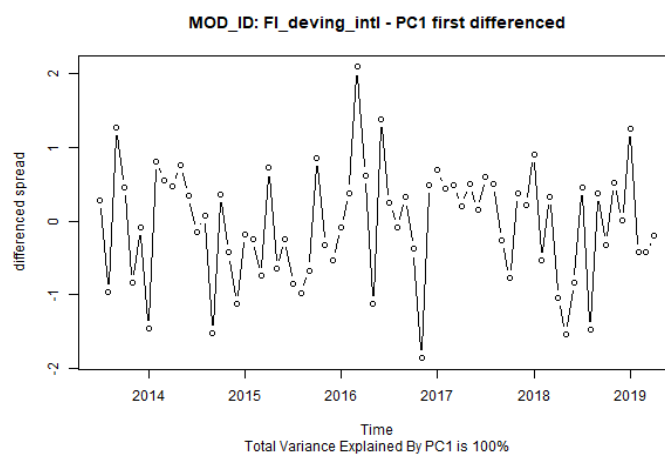
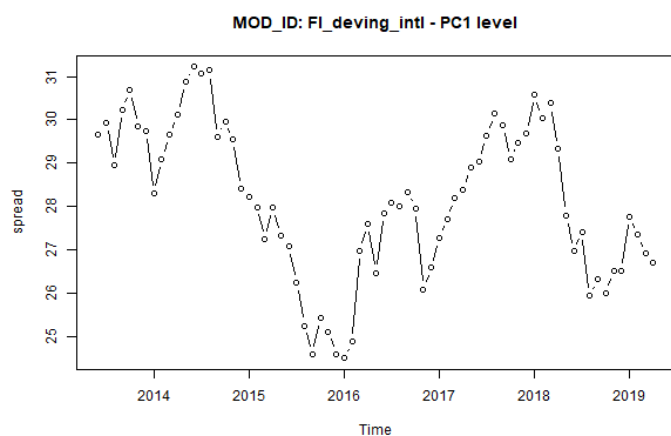
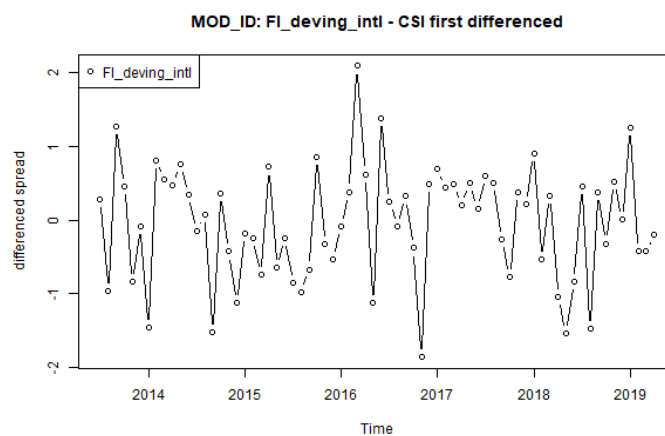
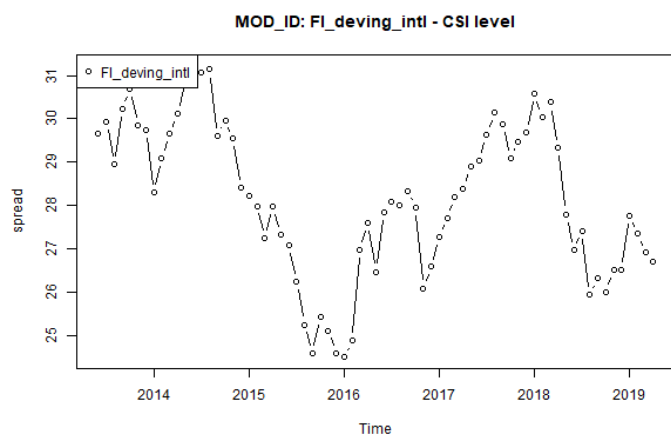
	EIGENVECTOR	MULTIPLIER
FI.deved.intl	1.00	1.00

Kendall Correlation

	FI_deved_intl	pc1
FI_deved_intl	1	1
pc1	1	1

Spearman Correlation

	FI_deved_intl	pc1
FI_deved_intl	1	1
pc1	1	1

1.8 Cluster: FI_{deving_intl} 

	EIGENVECTOR	MULTIPLIER
FI.deving_intl	1.00	1.00

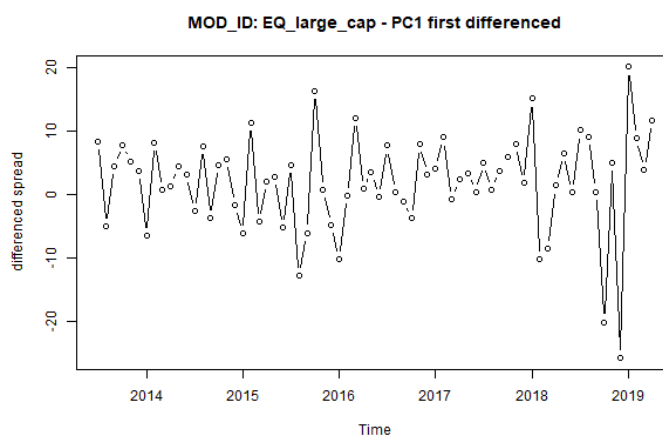
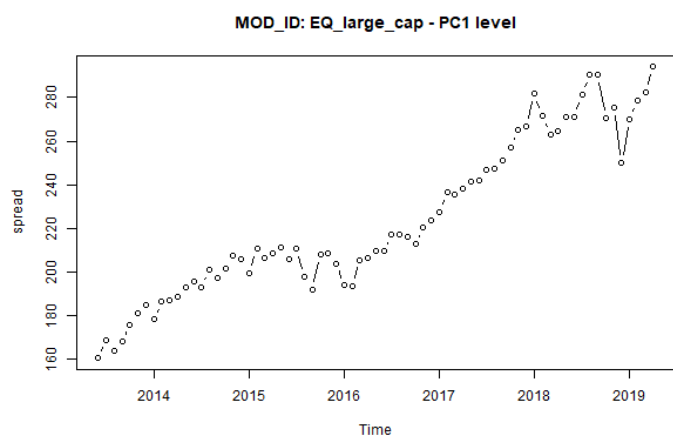
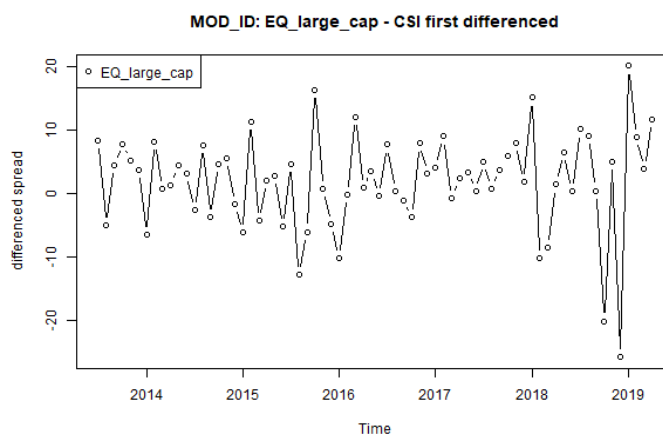
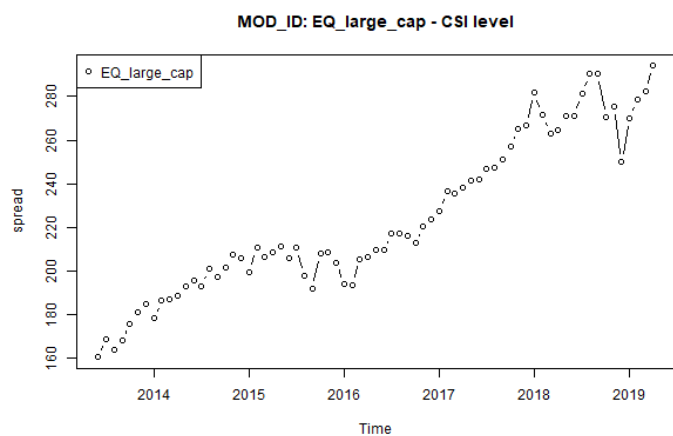
Kendall Correlation

	FI_deving_intl	pc1
FI_deving_intl	1	1
pc1	1	1

Spearman Correlation

	FI_deving_intl	pc1
FI_deving_intl	1	1
pc1	1	1

1.9 Cluster: EQ_{large_cap}



Total Variance Explained By PC1 is 100%

	EIGENVECTOR	MULTIPLIER
EQ_large_cap	1.00	1.00

Kendall Correlation

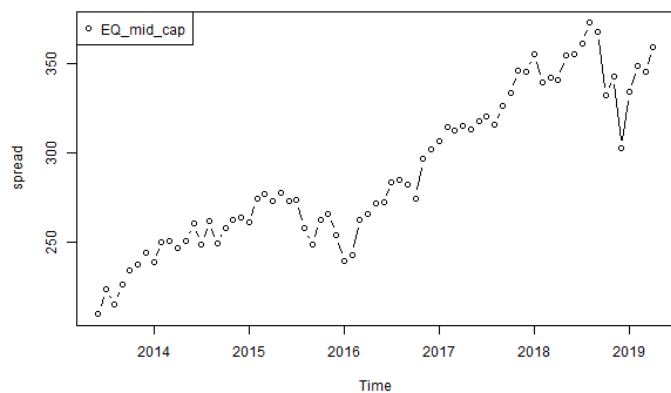
	EQ_large_cap	pc1
EQ_large_cap	1	1
pc1	1	1

Spearman Correlation

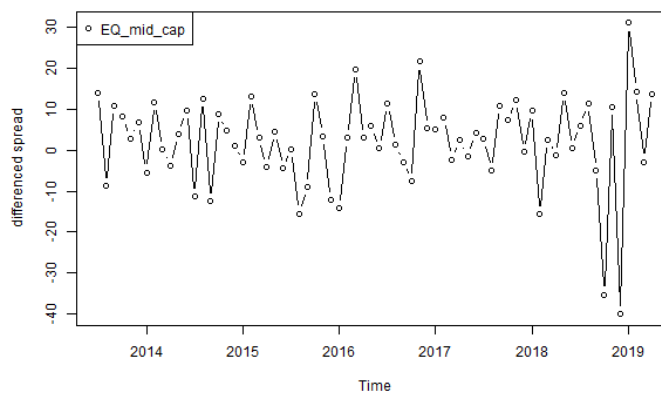
	EQ_large_cap	pc1
EQ_large_cap	1	1
pc1	1	1

1.10 Cluster: EQ_{mid_cap}

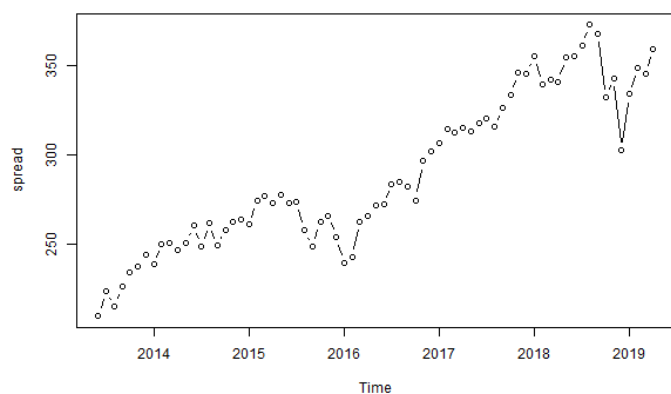
MOD_ID: EQ_mid_cap - CSI level



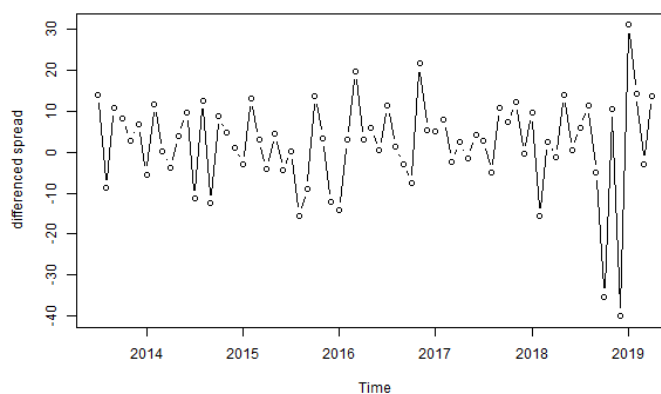
MOD_ID: EQ_mid_cap - CSI first differenced



MOD_ID: EQ_mid_cap - PC1 level



MOD_ID: EQ_mid_cap - PC1 first differenced



Total Variance Explained By PC1 is 100%

	EIGENVECTOR	MULTIPLIER
EQ_mid_cap	1.00	1.00

Kendall Correlation

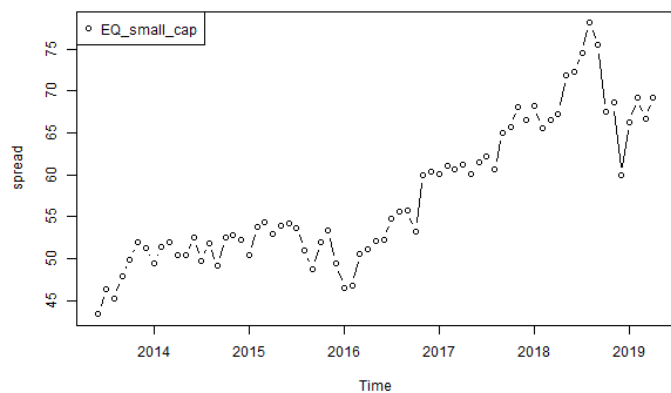
	EQ_mid_cap	pc1
EQ_mid_cap	1	1
pc1	1	1

Spearman Correlation

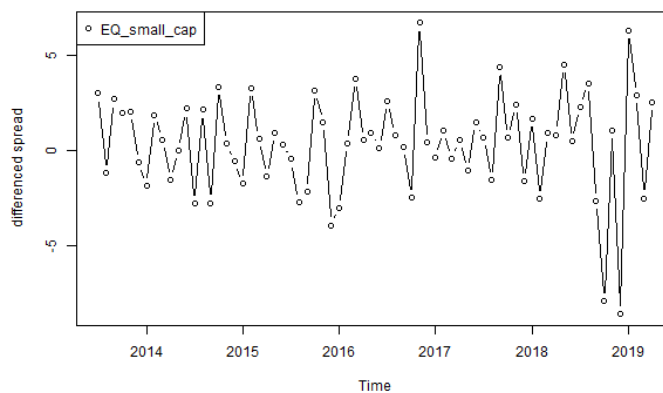
	EQ_mid_cap	pc1
EQ_mid_cap	1	1
pc1	1	1

1.11 Cluster: EQ_{small_cap}

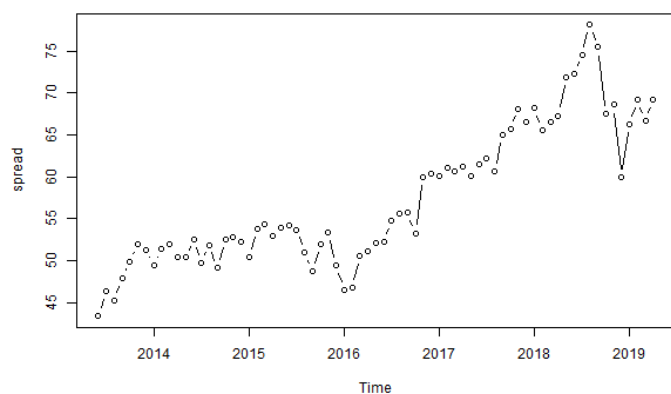
MOD_ID: EQ_small_cap - CSI level



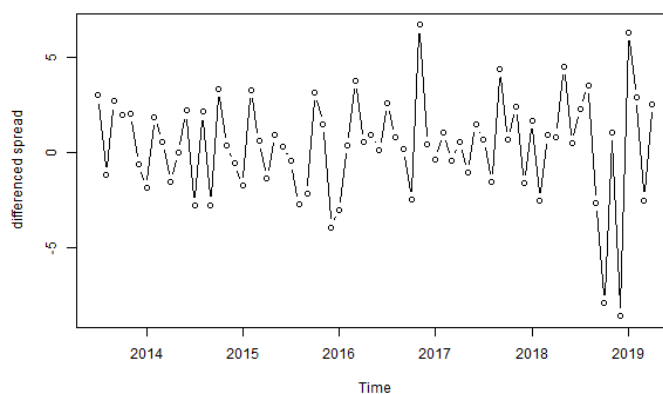
MOD_ID: EQ_small_cap - CSI first differenced



MOD_ID: EQ_small_cap - PC1 level



MOD_ID: EQ_small_cap - PC1 first differenced



Total Variance Explained By PC1 is 100%

	EIGENVECTOR	MULTIPLIER
EQ_small_cap	1.00	1.00

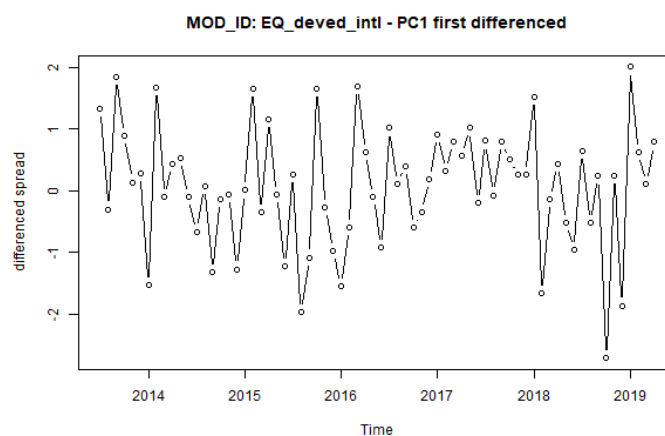
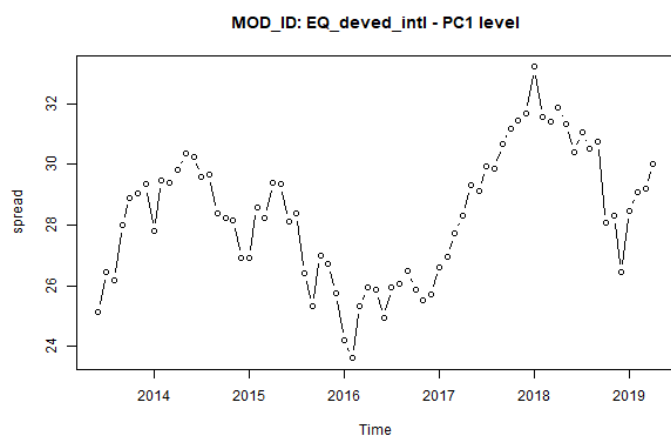
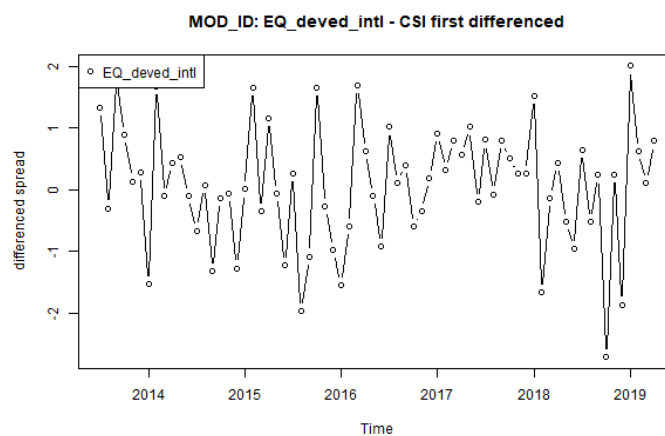
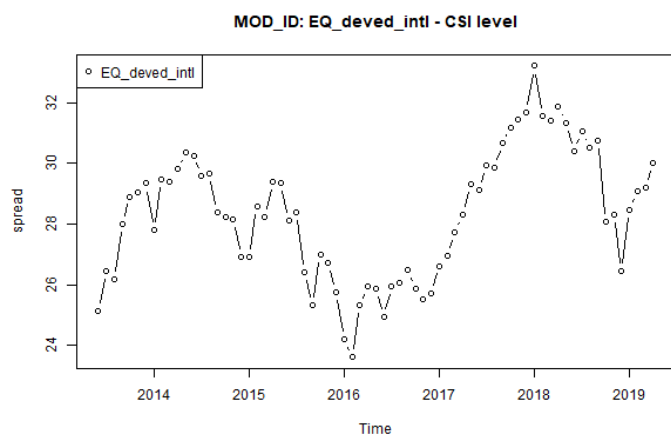
Kendall Correlation

	EQ_small_cap	pc1
EQ_small_cap	1	1
pc1	1	1

Spearman Correlation

	EQ_small_cap	pc1
EQ_small_cap	1	1
pc1	1	1

1.12 Cluster: EQ_{deved_intl}



Total Variance Explained By PC1 is 100%

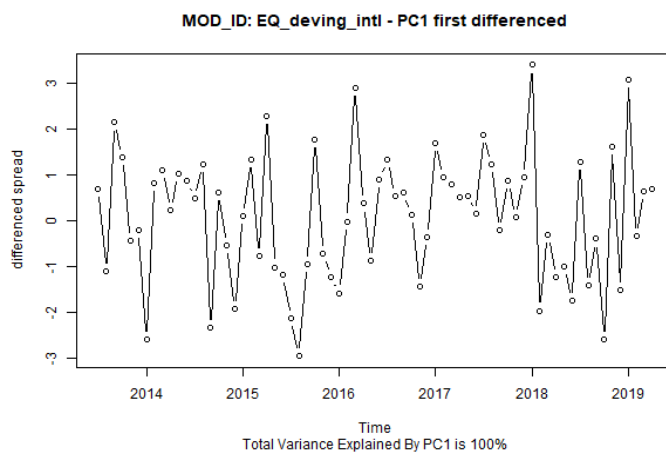
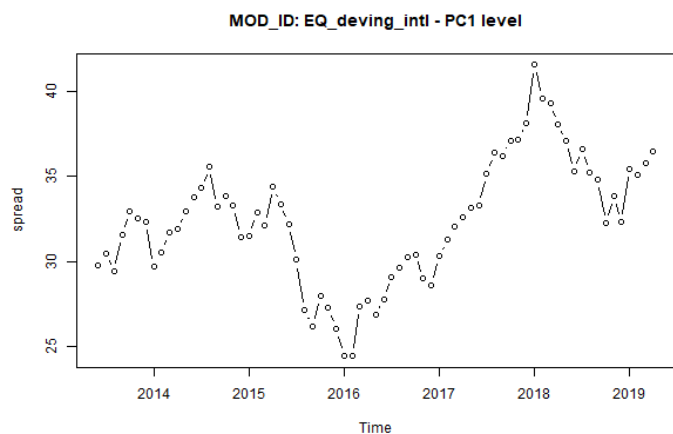
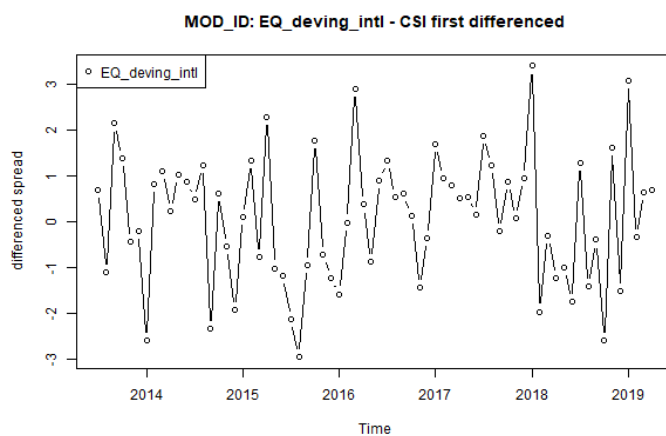
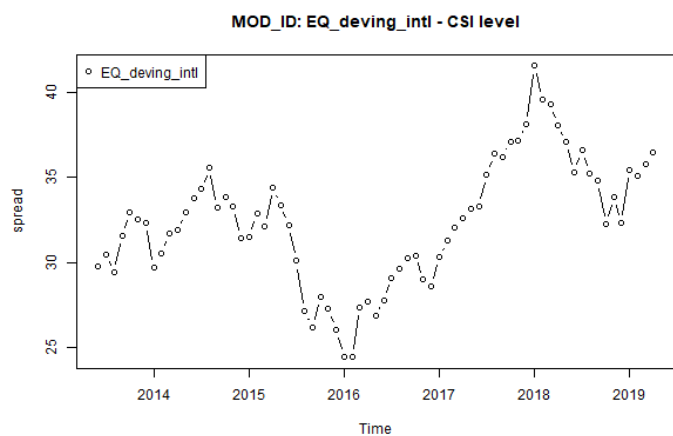
	EIGENVECTOR	MULTIPLIER
EQ_deved_intl	1.00	1.00

Kendall Correlation

	EQ_deved_intl	pc1
EQ_deved_intl	1	1
pc1	1	1

Spearman Correlation

	EQ_deved_intl	pc1
EQ_deved_intl	1	1
pc1	1	1

1.13 Cluster: EQ_{deving_intl} 

	EIGENVECTOR	MULTIPLIER
EQ_deving_intl	1.00	1.00

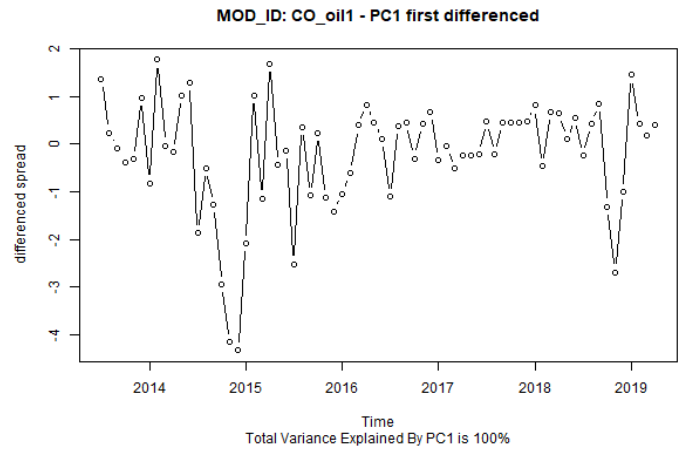
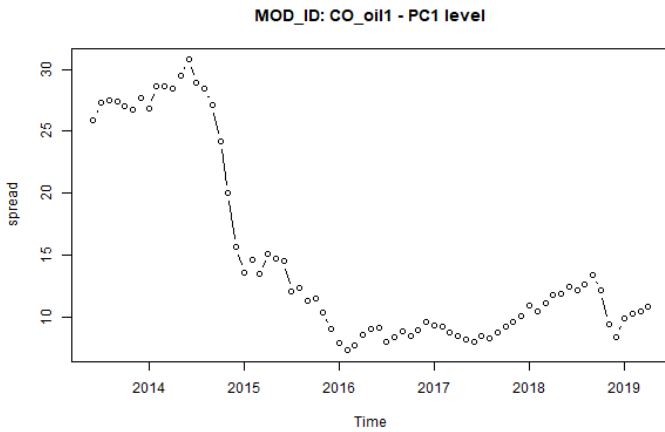
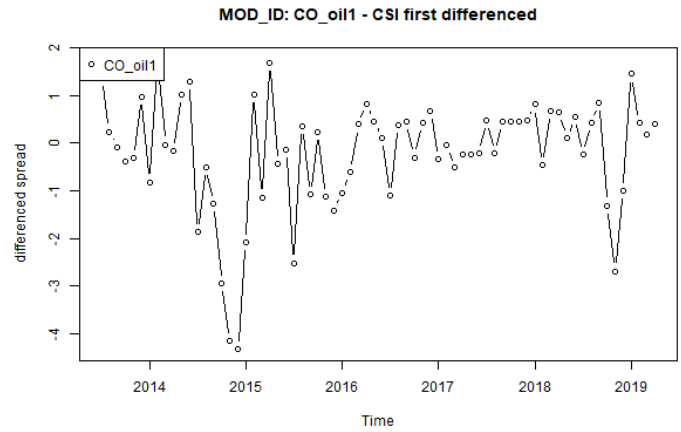
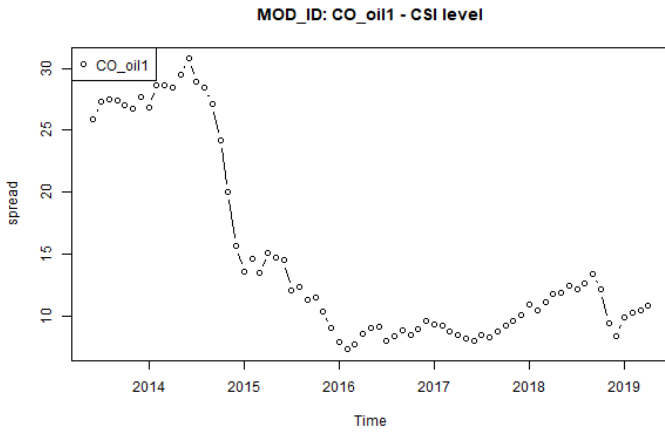
Kendall Correlation

	EQ_deving_intl	pc1
EQ_deving_intl	1	1
pc1	1	1

Spearman Correlation

	EQ_deving_intl	pc1
EQ_deving_intl	1	1
pc1	1	1

1.14 Cluster: CO_{oil1}



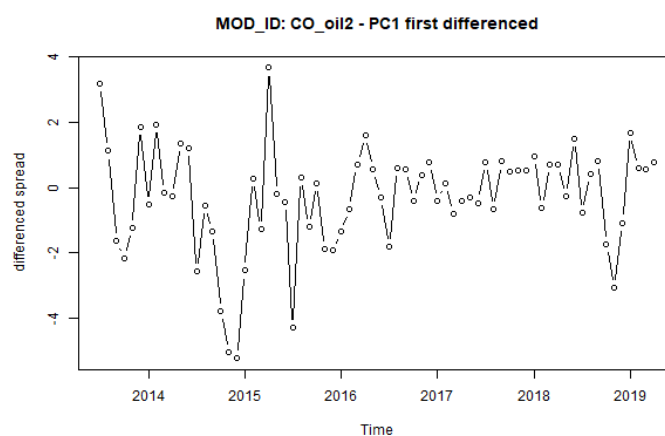
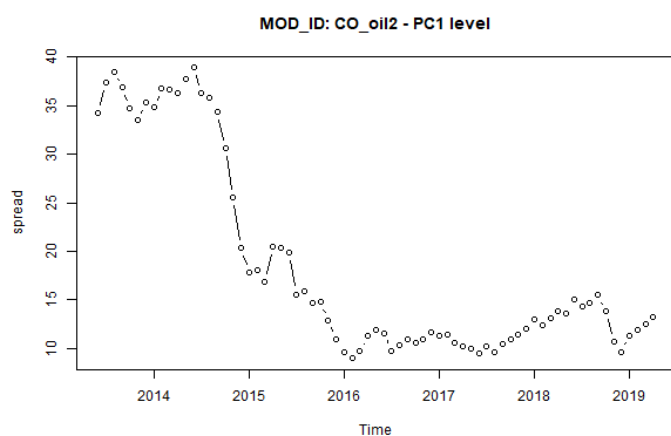
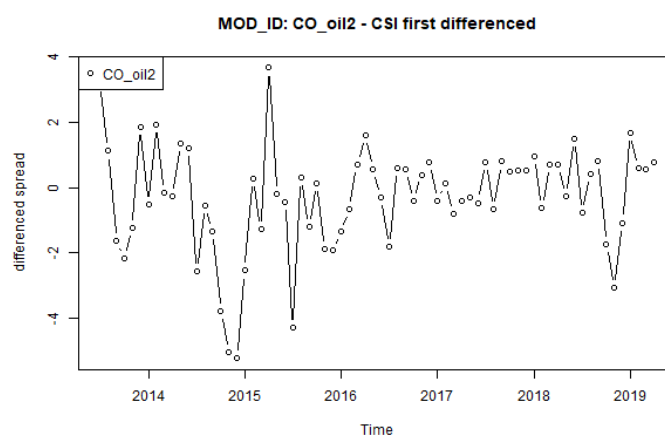
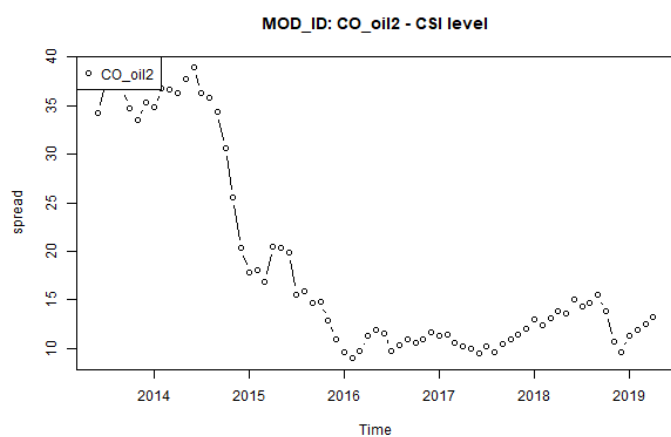
	EIGENVECTOR	MULTIPLIER
CO_oil1	1.00	1.00

Kendall Correlation

	CO_oil1	pc1
CO_oil1	1	1
pc1	1	1

Spearman Correlation

	CO_oil1	pc1
CO_oil1	1	1
pc1	1	1

1.15 Cluster: *CO_{oil2}*

Total Variance Explained By PC1 is 100%

	EIGENVECTOR	MULTIPLIER
CO_oil2	1.00	1.00

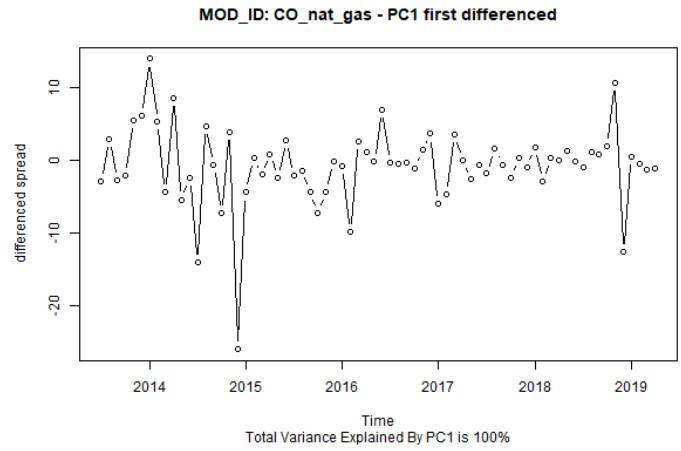
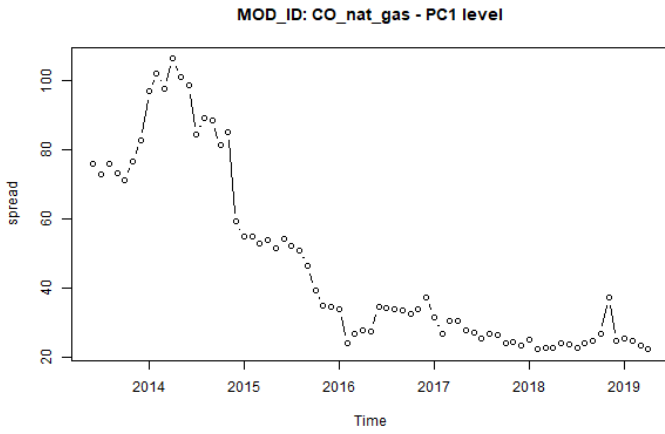
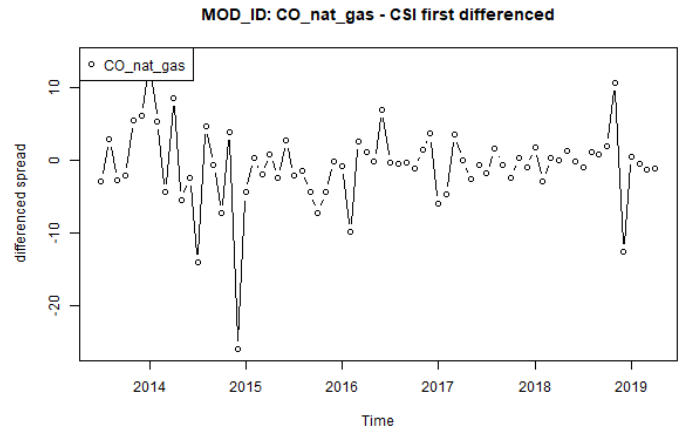
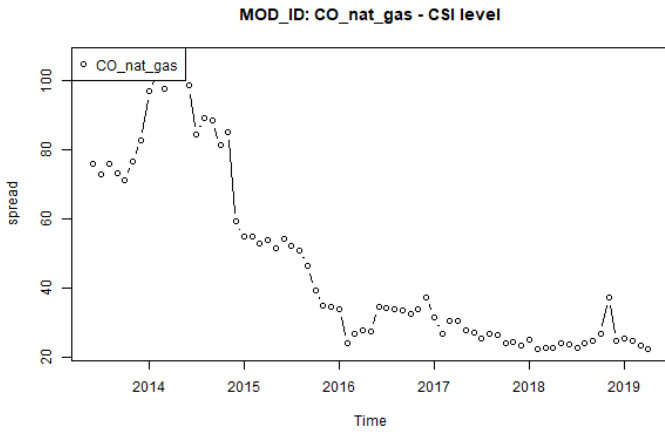
Kendall Correlation

	CO_oil2	pc1
CO_oil2	1	1
pc1	1	1

Spearman Correlation

	CO_oil2	pc1
CO_oil2	1	1
pc1	1	1

1.16 Cluster: CO_{nat_gas}



	EIGENVECTOR	MULTIPLIER
CO_nat_gas	1.00	1.00

Kendall Correlation

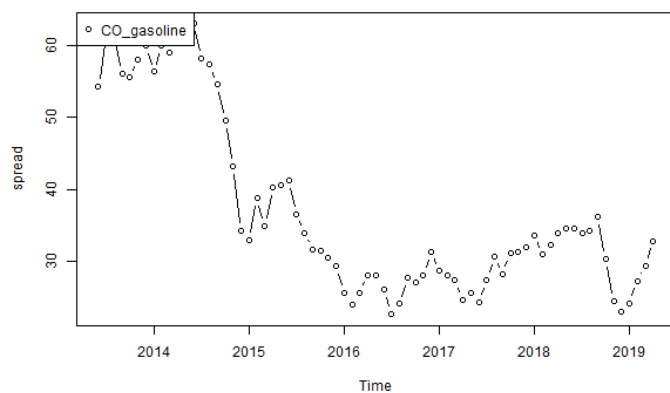
	CO_nat_gas	pc1
CO_nat_gas	1	1
pc1	1	1

Spearman Correlation

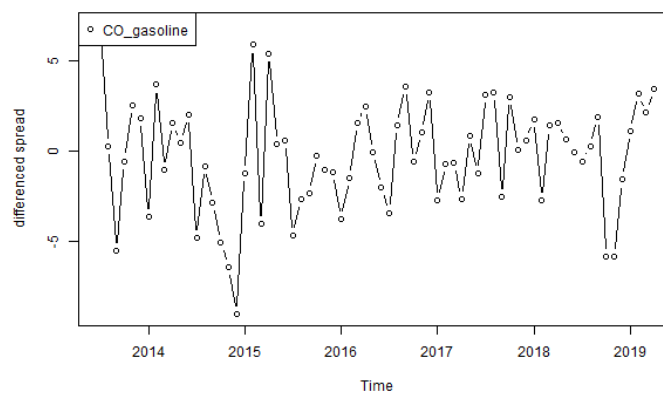
	CO_nat_gas	pc1
CO_nat_gas	1	1
pc1	1	1

1.17 Cluster: $CO_{gasoline}$

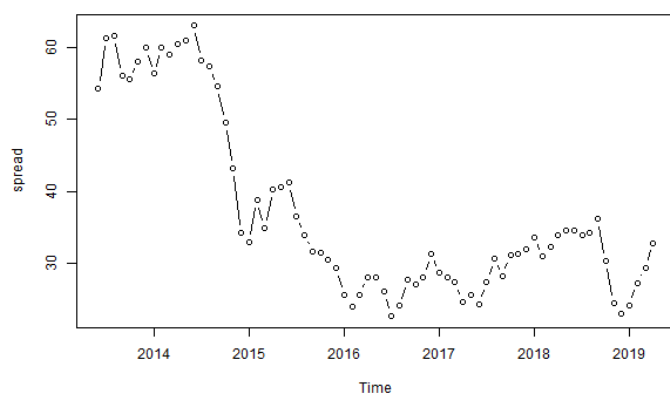
MOD_ID: CO_gasoline - CSI level



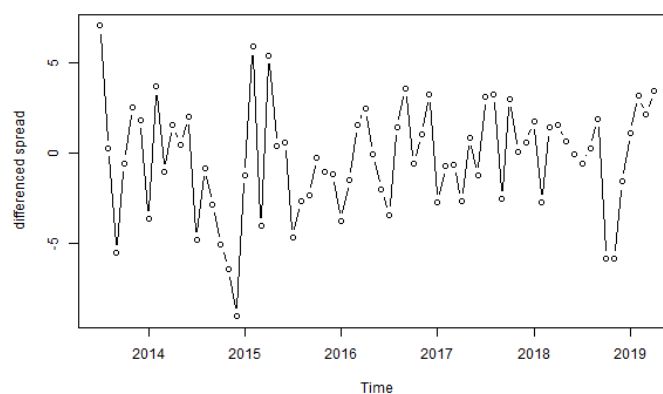
MOD_ID: CO_gasoline - CSI first differenced



MOD_ID: CO_gasoline - PC1 level



MOD_ID: CO_gasoline - PC1 first differenced



Total Variance Explained By PC1 is 100%

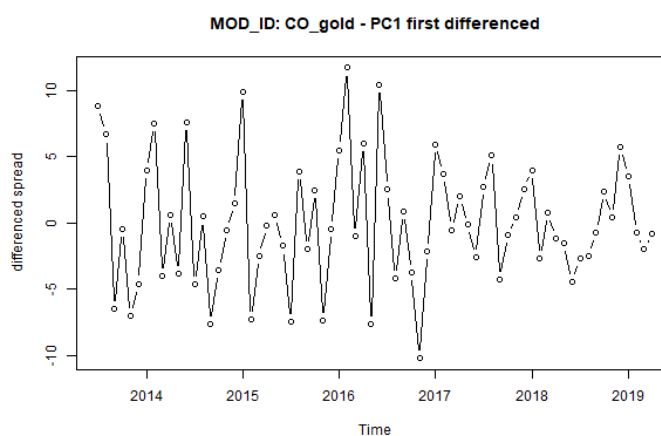
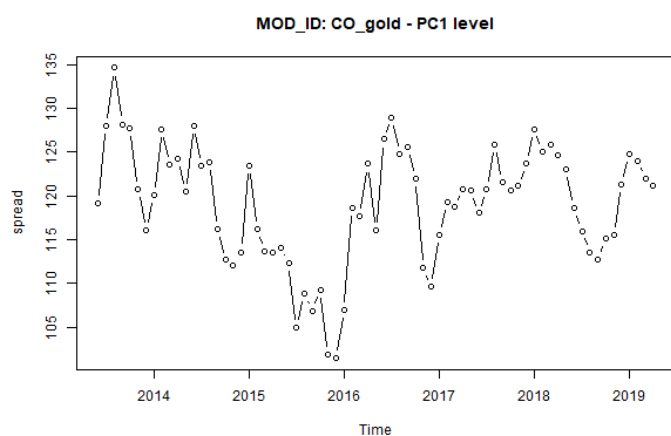
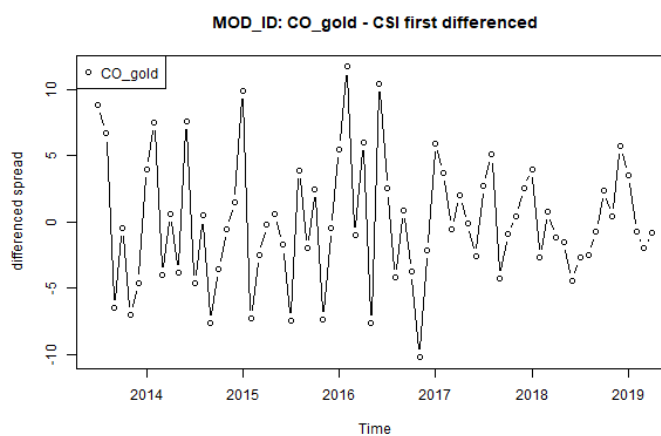
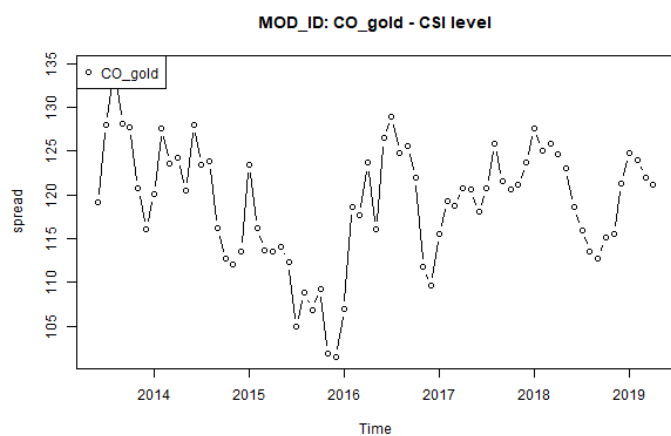
	EIGENVECTOR	MULTIPLIER
CO_gasoline	1.00	1.00

Kendall Correlation

	CO_gasoline	pc1
CO_gasoline	1	1
pc1	1	1

Spearman Correlation

	CO_gasoline	pc1
CO_gasoline	1	1
pc1	1	1

1.18 Cluster: CO_{gold} 

Total Variance Explained By PC1 is 100%

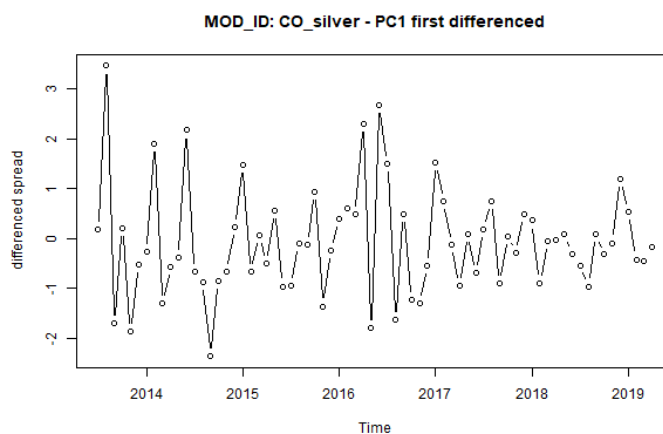
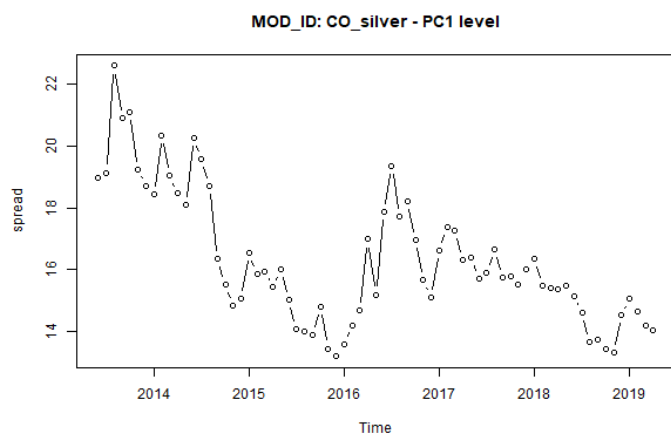
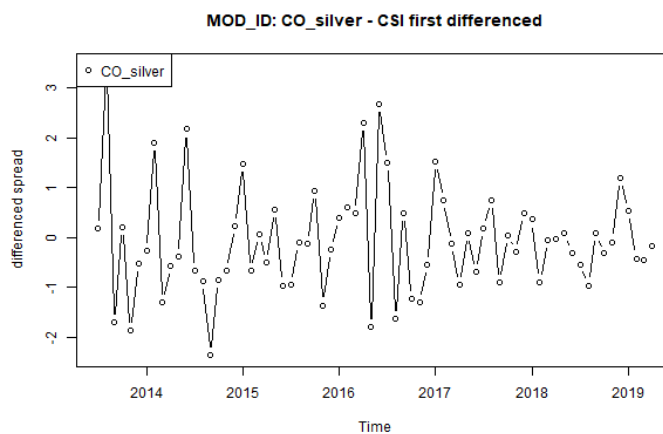
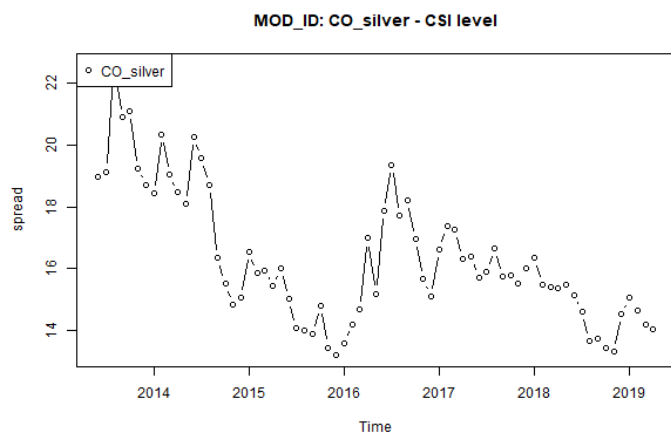
	EIGENVECTOR	MULTIPLIER
CO_gold	1.00	1.00

Kendall Correlation

	CO_gold	pc1
CO_gold	1	1
pc1	1	1

Spearman Correlation

	CO_gold	pc1
CO_gold	1	1
pc1	1	1

1.19 Cluster: *CO_silver*

Total Variance Explained By PC1 is 100%

	EIGENVECTOR	MULTIPLIER
CO_silver	1.00	1.00

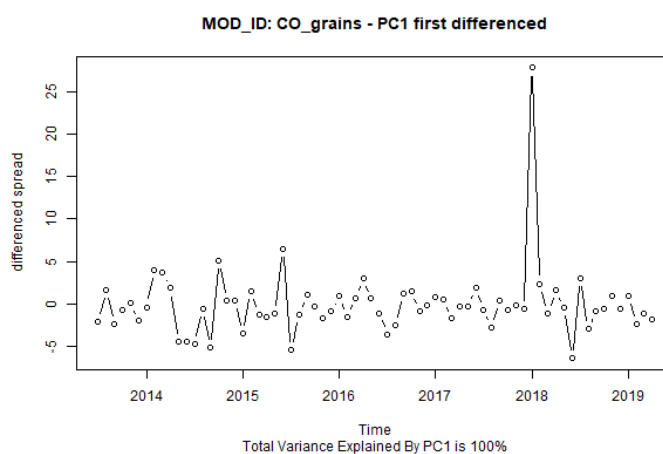
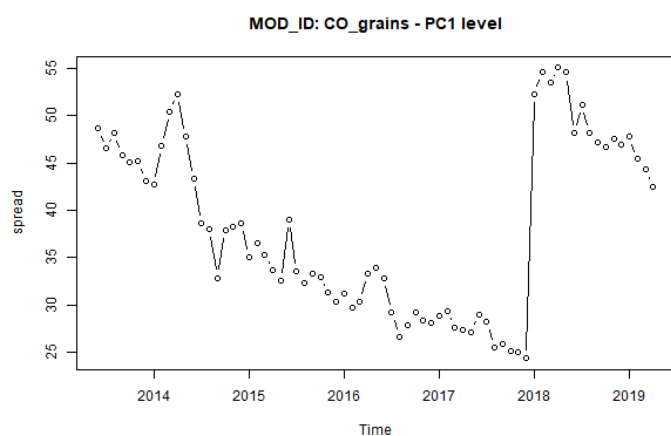
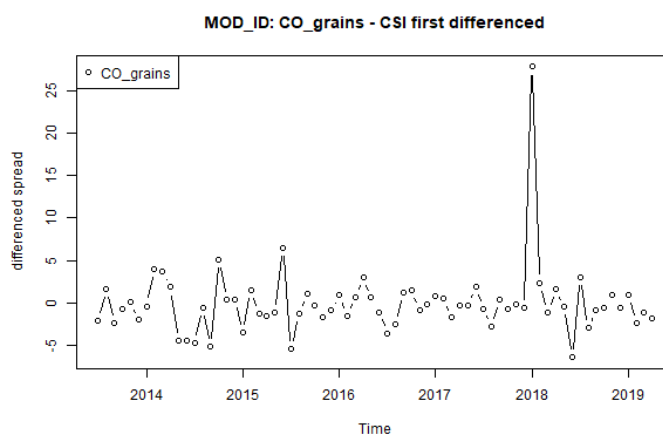
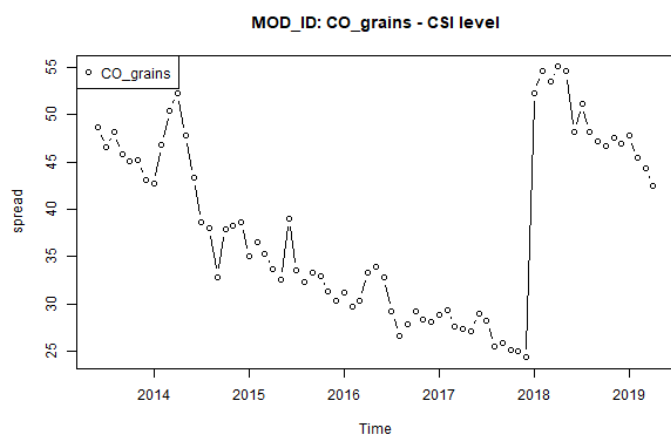
Kendall Correlation

	CO_silver	pc1
CO_silver	1	1
pc1	1	1

Spearman Correlation

	CO_silver	pc1
CO_silver	1	1
pc1	1	1

1.20 Cluster: CO_{grains}



	EIGENVECTOR	MULTIPLIER
CO_grains	1.00	1.00

Kendall Correlation

	CO_grains	pc1
CO_grains	1	1
pc1	1	1

Spearman Correlation

	CO_grains	pc1
CO_grains	1	1
pc1	1	1